

Optimization Lecture 11: Interior-point methods

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Outline

Inequality constrained minimization

Logarithmic barrier and central path

Barrier method

Phase I methods

Complexity analysis

Generalized inequalities

Inequality constrained minimization

Inequality constrained minimization

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

we assume

- ▶ f_i convex, twice continuously differentiable
- ▶ $A \in \mathbf{R}^{p \times n}$ with $\text{rank } A = p$
- ▶ p^* is finite and attained
- ▶ problem is strictly feasible: there exists $\tilde{x}$ with

$$\tilde{x} \in \text{dom } f_0, \quad f_i(\tilde{x}) < 0, \quad i = 1, \dots, m, \quad A\tilde{x} = b$$

hence, strong duality holds and dual optimum is attained

Examples

- ▶ LP, QP, QCQP, GP
- ▶ entropy maximization with linear inequality constraints

$$\begin{array}{ll}\text{minimize} & \sum_{i=1}^n x_i \log x_i \\ \text{subject to} & Fx \leq g, \quad Ax = b\end{array}$$

with $\text{dom } f_0 = \mathbf{R}_{++}^n$

- ▶ differentiability may require reformulating the problem, e.g., piecewise-linear minimization or ℓ_∞ -norm approximation via LP

Logarithmic barrier and central path

Logarithmic barrier

- ▶ reformulation via **indicator function**:

$$\begin{array}{ll}\text{minimize} & f_0(x) + \sum_{i=1}^m l_-(f_i(x)) \\ \text{subject to} & Ax = b\end{array}$$

where $l_-(u) = 0$ if $u \leq 0$, $l_-(u) = \infty$ otherwise

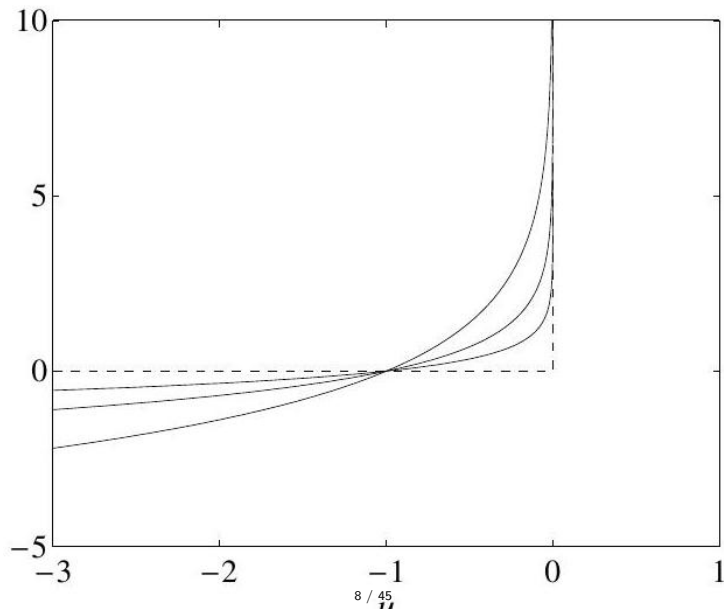
- ▶ **approximation via logarithmic barrier**:

$$\begin{array}{ll}\text{minimize} & f_0(x) - (1/t) \sum_{i=1}^m \log(-f_i(x)) \\ \text{subject to} & Ax = b\end{array}$$

- ▶ an equality constrained problem
- ▶ for $t > 0$, $-(1/t) \log(-u)$ is a smooth approximation of l_-
- ▶ approximation improves as $t \rightarrow \infty$

Logarithmic barrier

- $-(1/t) \log u$ for three values of t , and $I_-(u)$



Logarithmic barrier function

- ▶ log barrier function for constraints $f_1(x) \leq 0, \dots, f_m(x) \leq 0$

$$\phi(x) = - \sum_{i=1}^m \log(-f_i(x)), \quad \text{dom } \phi = \{x \mid f_1(x) < 0, \dots, f_m(x) < 0\}$$

- ▶ convex (from composition rules)
- ▶ twice continuously differentiable, with derivatives

$$\nabla \phi(x) = \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla f_i(x)$$

$$\nabla^2 \phi(x) = \sum_{i=1}^m \frac{1}{f_i(x)^2} \nabla f_i(x) \nabla f_i(x)^T + \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla^2 f_i(x)$$

Central path

- ▶ for $t > 0$, define $x^*(t)$ as the solution of

$$\begin{array}{ll}\text{minimize} & tf_0(x) + \phi(x) \\ \text{subject to} & Ax = b\end{array}$$

(for now, assume $x^*(t)$ exists and is unique for each $t > 0$)

- ▶ central path is $\{x^*(t) \mid t > 0\}$

example: central path for an LP

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, 6\end{array}$$

hyperplane $c^T x = c^T x^*(t)$ is tangent to level curve of ϕ through $x^*(t)$

Dual points on central path

- ▶ $x = x^*(t)$ if there exists a w such that

$$t \nabla f_0(x) + \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla f_i(x) + A^T w = 0, \quad Ax = b$$

- ▶ therefore, $x^*(t)$ minimizes the Lagrangian

$$L(x, \lambda^*(t), v^*(t)) = f_0(x) + \sum_{i=1}^m \lambda_i^*(t) f_i(x) + v^*(t)^T (Ax - b)$$

where we define $\lambda_i^*(t) = 1/(-t f_i(x^*(t)))$ and $v^*(t) = w/t$

- ▶ this confirms the intuitive idea that $f_0(x^*(t)) \rightarrow p^*$ if $t \rightarrow \infty$:

$$p^* \geq g(\lambda^*(t), v^*(t)) = L(x^*(t), \lambda^*(t), v^*(t)) = f_0(x^*(t)) - m/t$$

Interpretation via KKT conditions

$x = x^*(t)$, $\lambda = \lambda^*(t)$, $v = v^*(t)$ satisfy

1. primal constraints: $f_i(x) \leq 0, i = 1, \dots, m, Ax = b$
2. dual constraints: $\lambda \geq 0$
3. approximate complementary slackness: $-\lambda_i f_i(x) = 1/t, i = 1, \dots, m$
4. gradient of Lagrangian with respect to x vanishes:

$$\nabla f_0(x) + \sum_{i=1}^m \lambda_i \nabla f_i(x) + A^T v = 0$$

difference with KKT is that condition 3 replaces $\lambda_i f_i(x) = 0$

Force field interpretation

- ▶ centering problem (for problem with no equality constraints)

$$\text{minimize } tf_0(x) - \sum_{i=1}^m \log(-f_i(x))$$

- ▶ force field interpretation
 - ▶ $tf_0(x)$ is potential of force field $F_0(x) = -t\nabla f_0(x)$
 - ▶ $-\log(-f_i(x))$ is potential of force field $F_i(x) = (1/f_i(x)) \nabla f_i(x)$
- ▶ forces balance at $x^*(t)$:

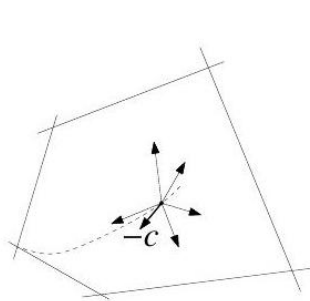
$$F_0(x^*(t)) + \sum_{i=1}^m F_i(x^*(t)) = 0$$

Example: LP

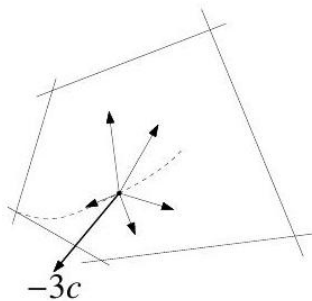
- ▶ minimize $c^T x$ subject to $a_i^T x \leq b_i, i = 1, \dots, m$, with $x \in \mathbf{R}^n$
- ▶ objective force field is constant: $F_0(x) = -tc$
- ▶ constraint force field decays as inverse distance to constraint hyperplane:

$$F_i(x) = \frac{-a_i}{b_i - a_i^T x}, \quad \|F_i(x)\|_2 = \frac{1}{\text{dist}(x, \mathcal{H}_i)}$$

where $\mathcal{H}_i = \{x \mid a_i^T x = b_i\}$



(a) $t = 1$



(b) $t = 3$

Barrier method

Barrier method

given strictly feasible x , $t := t^{(0)} > 0$, $\mu > 1$, tolerance $\epsilon > 0$.

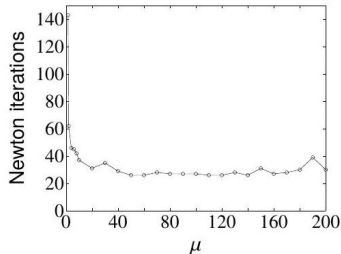
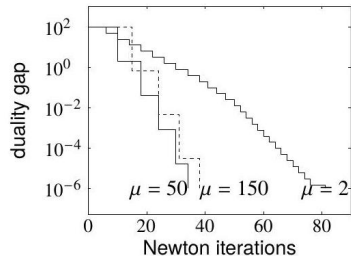
repeat

1. Centering step. Compute $x^*(t)$ by minimizing $tf_0 + \phi$, subject to $Ax = b$.
2. Update. $x := x^*(t)$.
3. Stopping criterion. **quit** if $m/t < \epsilon$.
4. Increase t . $t := \mu t$.

-
- ▶ terminates with $f_0(x) - p^* \leq \epsilon$ (stopping criterion follows from $f_0(x^*(t)) - p^* \leq m/t$)
 - ▶ centering usually done using Newton's method, starting at current x
 - ▶ choice of μ involves a trade-off: large μ means fewer outer iterations, more inner (Newton) iterations; typical values: $\mu = 10$ or 20
 - ▶ several heuristics for choice of $t^{(0)}$

Example: Inequality form LP

($m = 100$ inequalities, $n = 50$ variables)

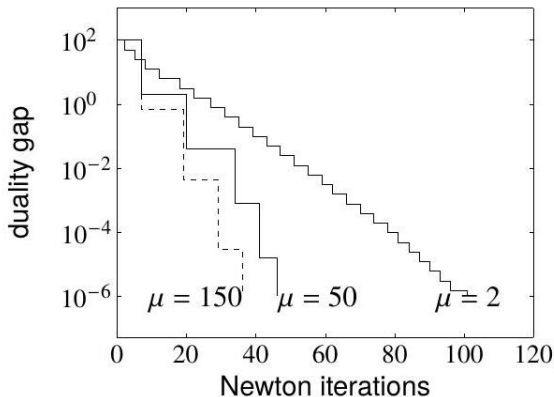


- ▶ starts with x on central path ($t^{(0)} = 1$, duality gap 100)
- ▶ terminates when $t = 10^8$ (gap 10^{-6})
- ▶ total number of Newton iterations not very sensitive for $\mu \geq 10$

Example: Geometric program in convex form

($m = 100$ inequalities and $n = 50$ variables)

$$\begin{array}{ll} \text{minimize} & \log \left(\sum_{k=1}^5 \exp(a_{0k}^T x + b_{0k}) \right) \\ \text{subject to} & \log \left(\sum_{k=1}^5 \exp(a_{ik}^T x + b_{ik}) \right) \leq 0, \quad i = 1, \dots, m \end{array}$$

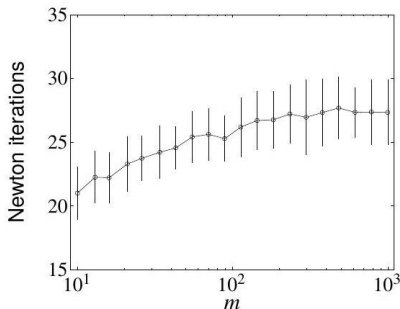


Family of standard LPs

$$(A \in \mathbf{R}^{m \times 2m})$$

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \geq 0\end{array}$$

$m = 10, \dots, 1000$; for each m , solve 100 randomly generated instances



number of iterations grows very slowly as m ranges over a 100 : 1 ratio

Phase I methods

Phase I methods

- ▶ barrier method needs strictly feasible starting point, i.e., x with

$$f_i(x) < 0, \quad i = 1, \dots, m, \quad Ax = b$$

- ▶ (like the infeasible start Newton method, more sophisticated interior-point methods do not require a feasible starting point)
- ▶ **phase I** method forms an optimization problem that
 - ▶ is itself strictly feasible
 - ▶ finds a strictly feasible point for original problem, if one exists
 - ▶ certifies original problem as infeasible otherwise
- ▶ **phase II** uses barrier method starting from strictly feasible point found in phase I

Basic phase I method

- ▶ introduce slack variable s in **phase I problem**

$$\begin{array}{ll}\text{minimize (over } x, s) & s \\ \text{subject to} & f_i(x) \leq s, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

with optimal value $\bar{p}^*$

- ▶ if $\bar{p}^* < 0$, original inequalities are strictly feasible
 - ▶ if $\bar{p}^* > 0$, original inequalities are infeasible
 - ▶ $\bar{p}^* = 0$ is an ambiguous case
- ▶ start phase I problem with
 - ▶ any $\tilde{x}$ in problem domain with $A\tilde{x} = b$
 - ▶ $s = 1 + \max_i f_i(\tilde{x})$

Sum of infeasibilities phase I method

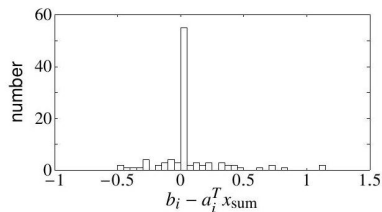
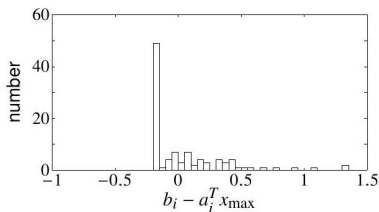
- ▶ minimize **sum** of slacks, not max:

$$\begin{array}{ll}\text{minimize} & \mathbf{1}^T s \\ \text{subject to} & s \geq 0, \quad f_i(x) \leq s_i, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

- ▶ for infeasible problems, produces a solution that satisfies many (but not all) inequalities
- ▶ can weight slacks to set **priorities** (in satisfying constraints)

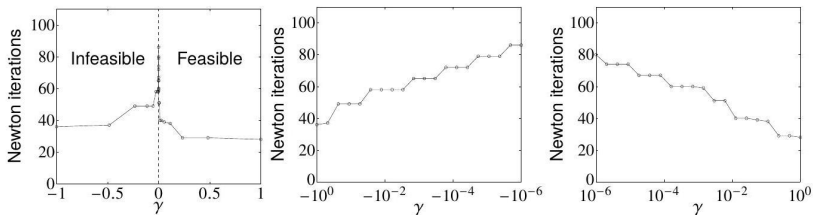
Example

- ▶ infeasible set of 100 linear inequalities in 50 variables
- ▶ left: basic phase I solution; satisfies 39 inequalities
- ▶ right: sum of infeasibilities phase I solution; satisfies 79 inequalities



Example: Family of linear inequalities

- ▶ $Ax \leq b + \gamma \Delta b$; strictly feasible for $\gamma > 0$, infeasible for $\gamma < 0$
- ▶ use basic phase I, terminate when $s < 0$ or dual objective is positive
- ▶ number of iterations roughly proportional to $\log(1/|\gamma|)$



Complexity analysis

Number of outer iterations

- ▶ in each iteration duality gap is reduced by exactly the factor μ
- ▶ **number of outer (centering) iterations** is exactly

$$\left\lceil \frac{\log(m/(\epsilon t^{(0)}))}{\log \mu} \right\rceil$$

plus the initial centering step (to compute $x^*(t^{(0)})$)

- ▶ we will bound **number of Newton steps per centering iteration** using self-concordance analysis

Complexity analysis via self-concordance

same assumptions as on slide 4, plus:

- ▶ sublevel sets (of f_0 , on the feasible set) are bounded
- ▶ $tf_0 + \phi$ is self-concordant with closed sublevel sets

Second condition

- ▶ holds for LP, QP, QCQP
- ▶ may require reformulating the problem, e.g.,

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^n x_i \log x_i \\ \text{subject to} & Fx \leq g \end{array} \quad \longrightarrow \quad \begin{array}{ll} \text{minimize} & \sum_{i=1}^n x_i \log x_i \\ \text{subject to} & Fx \leq g, \quad x \geq 0 \end{array}$$

- ▶ needed for complexity analysis; barrier method works even when self-concordance assumption does not apply

Newton iterations per centering step

- ▶ we compute $x^+ = x^*(\mu t)$, by minimizing $\mu t f_0(x) + \phi(x)$ starting from $x = x^*(t)$
- ▶ from self-concordance theory,

$$\# \text{Newton iterations} \leq \frac{\mu t f_0(x) + \phi(x) - \mu t f_0(x^+) - \phi(x^+)}{\gamma} + c$$

- ▶ γ, c are constants (that depend only on Newton algorithm parameters)

Newton iterations per centering step

- ▶ we will bound numerator $\mu t f_0(x) + \phi(x) - \mu t f_0(x^+) - \phi(x^+)$
- ▶ with $\lambda_i = \lambda_i^*(t) = -1/(t f_i(x))$, we have $-f_i(x) = 1/(t \lambda_i)$, so

$$\phi(x) = \sum_{i=1}^m -\log(-f_i(x)) = \sum_{i=1}^m \log(t \lambda_i)$$

so

$$\begin{aligned}\phi(x) - \phi(x^+) &= \sum_{i=1}^m (\log(t \lambda_i) + \log(-f_i(x^+))) \\ &= \sum_{i=1}^m \log(-\mu t \lambda_i f_i(x^+)) - m \log \mu\end{aligned}$$

Newton iterations per centering step

using $\log u \leq u - 1$ we have

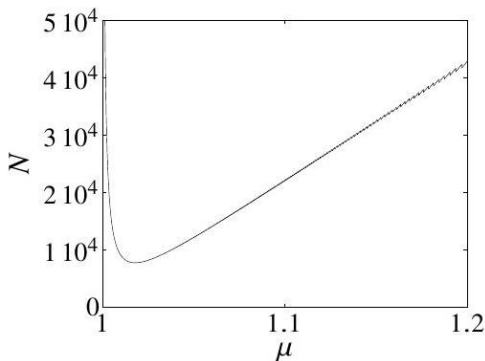
$$\phi(x) - \phi(x^+) \leq -\mu t \sum_{i=1}^m \lambda_i f_i(x^+) - m - m \log \mu, \text{ so}$$

$$\begin{aligned} & \mu t f_0(x) + \phi(x) - \mu t f_0(x^+) - \phi(x^+) \\ & \leq \mu t f_0(x) - \mu t f_0(x^+) - \mu t \sum_{i=1}^m \lambda_i f_i(x^+) - m - m \log \mu \\ & = \mu t f_0(x) - \mu t \left(f_0(x^+) + \sum_{i=1}^m \lambda_i f_i(x^+) + v^T (Ax^+ - b) \right) - m - m \log \mu \\ & = \mu t f_0(x) - \mu t L(x^+, \lambda, v) - m - m \log \mu \\ & \leq \mu t f_0(x) - \mu t g(\lambda, v) - m - m \log \mu \\ & = m(\mu - 1 - \log \mu) \end{aligned}$$

using $L(x^+, \lambda, \mu t) \geq g(\lambda, v)$ in second last line and
 $f_0(x) - g(\lambda, v) = m/t$ in last line

Total number of Newton iterations

$$\# \text{Newton iterations} \leq N = \left\lceil \frac{\log(m/(t^{(0)}\epsilon))}{\log \mu} \right\rceil \left(\frac{m(\mu-1-\log \mu)}{\gamma} + c \right)$$



N versus μ for typical values of γ, c ; $m = 100$, initial duality gap $\frac{m}{t^{(0)}\epsilon} = 10^5$

- confirms trade-off in choice of μ
- in practice, #iterations is in the tens; not very sensitive for $\mu \geq 10$

Polynomial-time complexity of barrier method

- ▶ for $\mu = 1 + 1/\sqrt{m}$:

$$N = O\left(\sqrt{m} \log\left(\frac{m/t^{(0)}}{\epsilon}\right)\right)$$

- ▶ number of Newton iterations for fixed gap reduction is $O(\sqrt{m})$
- ▶ multiply with cost of one Newton iteration (a polynomial function of problem dimensions), to get bound on number of flops
- ▶ this choice of μ optimizes worst-case complexity; in practice we choose μ fixed and larger

Generalized inequalities

Generalized inequalities

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq_{K_i} 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

- ▶ f_0 convex, $f_i : \mathbf{R}^n \rightarrow \mathbf{R}^{k_i}, i = 1, \dots, m$, convex with respect to proper cones $K_i \in \mathbf{R}^{k_i}$
- ▶ we assume
 - ▶ f_i twice continuously differentiable
 - ▶ $A \in \mathbf{R}^{p \times n}$ with $\text{rank } A = p$
 - ▶ p^* is finite and attained
 - ▶ problem is strictly feasible; hence strong duality holds and dual optimum is attained
- ▶ examples of greatest interest: SOCP, SDP

Generalized logarithm for proper cone

$\psi : \mathbf{R}^q \rightarrow \mathbf{R}$ is **generalized logarithm** for proper cone $K \subseteq \mathbf{R}^q$ if:

- ▶ $\text{dom } \psi = \text{int } K$ and $\nabla^2 \psi(y) \prec 0$ for $y \succ_K 0$
- ▶ $\psi(sy) = \psi(y) + \theta \log s$ for $y \succ_K 0, s > 0$ (θ is the degree of ψ)

examples

- ▶ nonnegative orthant $K = \mathbf{R}_+^n : \psi(y) = \sum_{i=1}^n \log y_i$, with degree $\theta = n$
- ▶ positive semidefinite cone $K = \mathbf{S}_+^n : \psi(Y) = \log \det Y$, with degree $\theta = n$
- ▶ second-order cone $K = \left\{ y \in \mathbf{R}^{n+1} \mid (y_1^2 + \dots + y_n^2)^{1/2} \leq y_{n+1} \right\} :$

$$\psi(y) = \log (y_{n+1}^2 - y_1^2 - \dots - y_n^2) \quad \text{with degree } (\theta = 2)$$

Properties

- ▶ (without proof): for $y \succ_K 0$,

$$\nabla\psi(y) \succeq_{K^*} 0, \quad y^T \nabla\psi(y) = \theta$$

- ▶ nonnegative orthant $\mathbf{R}_+^n : \psi(y) = \sum_{i=1}^n \log y_i$

$$\nabla\psi(y) = (1/y_1, \dots, 1/y_n), \quad y^T \nabla\psi(y) = n$$

- ▶ positive semidefinite cone $\mathbf{S}_+^n : \psi(Y) = \log \det Y$

$$\nabla\psi(Y) = Y^{-1}, \quad \text{tr}(Y \nabla\psi(Y)) = n$$

- ▶ second-order cone $K = \left\{ y \in \mathbf{R}^{n+1} \mid (y_1^2 + \dots + y_n^2)^{1/2} \leq y_{n+1} \right\} :$

$$\nabla\psi(y) = \frac{2}{y_{n+1}^2 - y_1^2 - \dots - y_n^2} \begin{bmatrix} -y_1 \\ \vdots \\ -y_n \\ y_{n+1} \end{bmatrix}, \quad y^T \nabla\psi(y) = 2$$

Logarithmic barrier and central path

logarithmic barrier for $f_1(x) \leq_{K_1} 0, \dots, f_m(x) \leq_{K_m} 0$:

$$\phi(x) = - \sum_{i=1}^m \psi_i(-f_i(x)), \quad \text{dom } \phi = \{x \mid f_i(x) \prec_{K_i} 0, i = 1, \dots, m\}$$

- ▶ ψ_i is generalized logarithm for K_i , with degree θ_i
- ▶ ϕ is convex, twice continuously differentiable

central path: $\{x^*(t) \mid t > 0\}$ where $x^*(t)$ is solution of

$$\begin{array}{ll} \text{minimize} & tf_0(x) + \phi(x) \\ \text{subject to} & Ax = b \end{array}$$

Dual points on central path

$x = x^*(t)$ if there exists $w \in \mathbf{R}^p$,

$$t \nabla f_0(x) + \sum_{i=1}^m Df_i(x)^T \nabla \psi_i(-f_i(x)) + A^T w = 0$$

$(Df_i(x) \in \mathbf{R}^{k_i \times n}$ is derivative matrix of f_i)

► therefore, $x^*(t)$ minimizes Lagrangian $L(x, \lambda^*(t), v^*(t))$, where

$$\lambda_i^*(t) = \frac{1}{t} \nabla \psi_i(-f_i(x^*(t))), \quad v^*(t) = \frac{w}{t}$$

► from properties of $\psi_i : \lambda_i^*(t) \succ_{\kappa_i^*} 0$, with duality gap

$$f_0(x^*(t)) - g(\lambda^*(t), v^*(t)) = (1/t) \sum_{i=1}^m \theta_i$$

Example: Semidefinite programming

(with $F_i \in \mathbf{S}^p$)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & F(x) = \sum_{i=1}^n x_i F_i + G \preceq 0\end{array}$$

- ▶ logarithmic barrier: $\phi(x) = \log \det(-F(x)^{-1})$
- ▶ central path: $x^*(t)$ minimizes $tc^T x - \log \det(-F(x))$; hence

$$tc_i - \text{tr}\left(F_i F(x^*(t))^{-1}\right) = 0, \quad i = 1, \dots, n$$

- ▶ dual point on central path: $Z^*(t) = -(1/t)F(x^*(t))^{-1}$ is feasible for

$$\begin{array}{ll}\text{maximize} & \text{tr}(GZ) \\ \text{subject to} & \text{tr}(F_i Z) + c_i = 0, \quad i = 1, \dots, n \\ & Z \succeq 0\end{array}$$

- ▶ duality gap on central path: $c^T x^*(t) - \text{tr}(GZ^*(t)) = p/t$

Barrier method

given strictly feasible x , $t := t^{(0)} > 0$, $\mu > 1$, tolerance $\epsilon > 0$.

repeat

1. Centering step. Compute $x^*(t)$ by minimizing $tf_0 + \phi$, subject to $Ax = b$.
2. Update. $x := x^*(t)$.
3. Stopping criterion. quit if $(\sum_i \theta_i) / t < \epsilon$.
4. Increase t . $t := \mu t$.

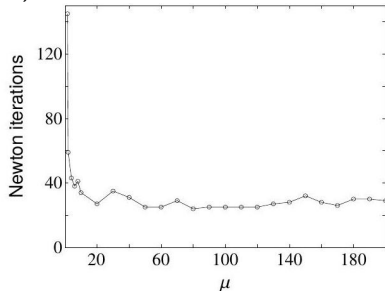
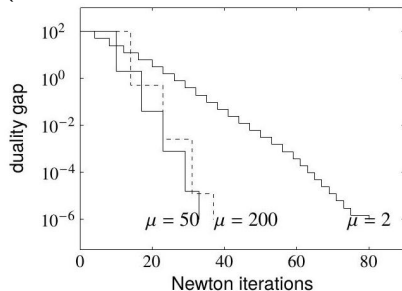
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- ▶ only difference is duality gap m/t on central path is replaced by $\sum_i \theta_i / t$
 - ▶ number of outer iterations:

$$\left\lceil \frac{\log((\sum_i \theta_i) / (\epsilon t^{(0)}))}{\log \mu} \right\rceil$$

- ▶ complexity analysis via self-concordance applies to SDP, SOCP

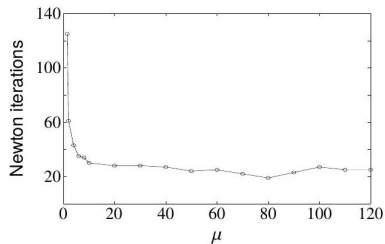
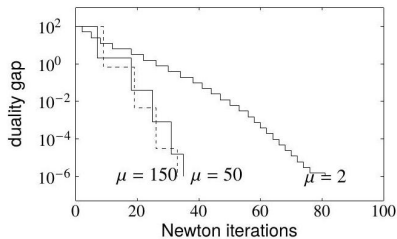
Example: SOCP

(50 variables, 50 SOC constraints in $\mathbf{R}^6$)



Example: SDP

(100 variables, LMI constraint in $\mathbf{S}^{100}$)

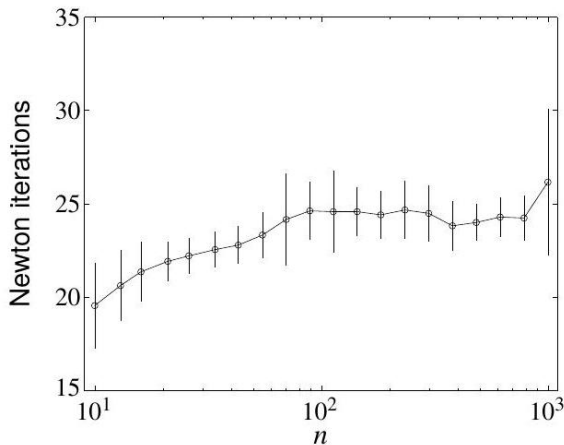


Example: Family of SDPs

$$(A \in \mathbf{S}^n, x \in \mathbf{R}^n)$$

$$\begin{array}{ll}\text{minimize} & \mathbf{1}^T x \\ \text{subject to} & A + \mathbf{diag}(x) \geq 0\end{array}$$

$n = 10, \dots, 1000$; for each n solve 100 randomly generated instances



Primal-dual interior-point methods

- ▶ more efficient than barrier method when high accuracy is needed
- ▶ update primal and dual variables, and κ , at each iteration; no distinction between inner and outer iterations
- ▶ often exhibit superlinear asymptotic convergence
- ▶ search directions can be interpreted as Newton directions for modified KKT conditions
- ▶ can start at infeasible points
- ▶ cost per iteration same as barrier method